

Cross-Platform ETL, Sentiment Analysis & Conversion Funnel Diagnostics

An end-to-end marketing analytics portfolio project spanning legacy data migration, NLP-driven sentiment scoring, and interactive conversion funnel diagnostics for e-commerce.

MARKETING ANALYTICS

E-COMMERCE

PYTHON · SQL · POWER BI

CONVERSION FUNNEL DIAGNOSTICS



Project Overview

Executive Summary

This project delivers a complete analytics workflow — from raw legacy data to actionable business intelligence. Three integrated phases uncover conversion bottlenecks, validate customer sentiment, and surface operational opportunities.

ETL Migration

Migrated a proprietary SQL Server .bak backup into a production-ready PostgreSQL environment via a custom Python pipeline.

Sentiment Scoring

Applied VADER NLP scoring to unstructured customer reviews, cross-referenced against star ratings to detect discrepancies.

Funnel Diagnostics

Built an interactive Power BI diagnostic layer to isolate friction points between content views, clicks, and transactions.

The Challenge

Business Problem & Objectives

Corporate acquisitions left critical marketing and event logs trapped in an isolated SQL Server backup. A stagnant conversion rate and unquantified sentiment swings demanded a unified analytical response.

Reconstruct Legacy Assets

Extract and port data from a proprietary .bak file to PostgreSQL without integrity loss.

Audit Data Integrity

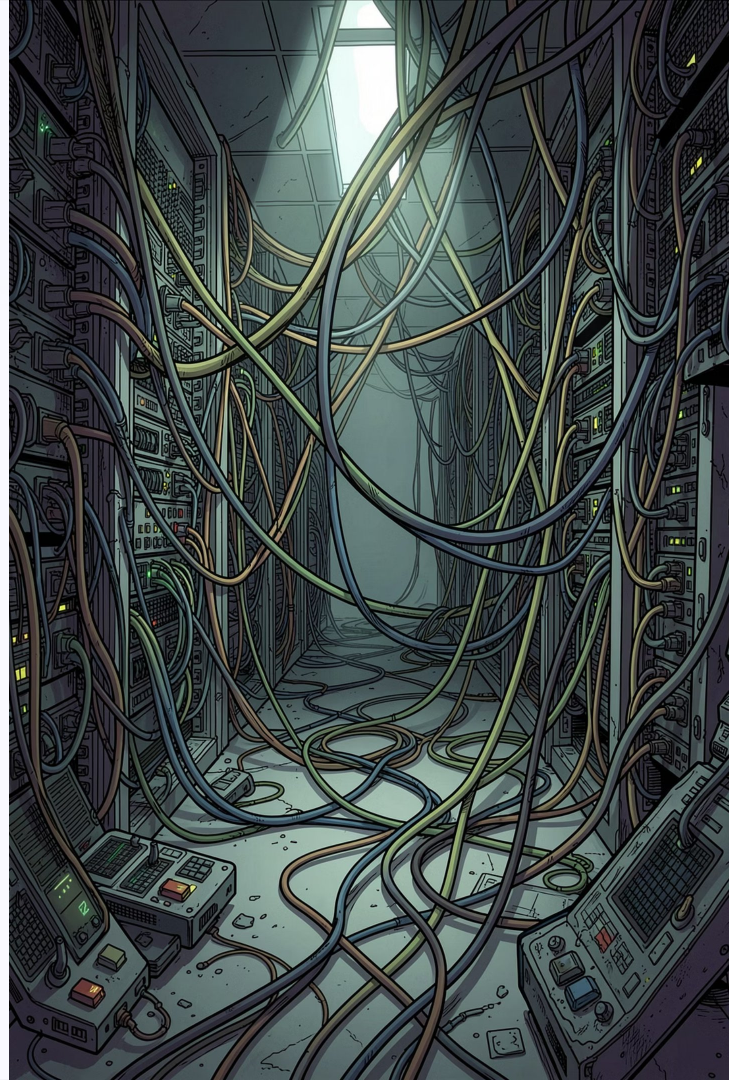
Isolate duplicated journey events that artificially inflate web traffic metrics.

Execute Sentiment Classification

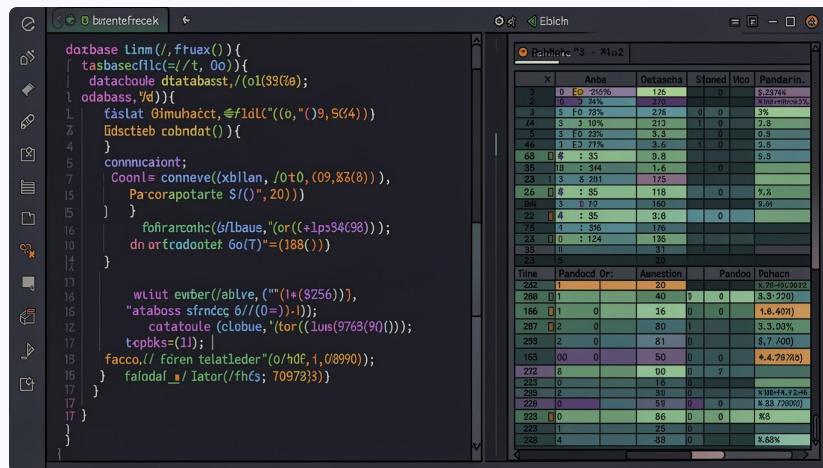
Automate NLP text processing to uncover gaps between review text and star ratings.

Pinpoint Funnel Attrition

Measure micro-conversion drops between views, click-throughs, and final transactions.



Cross-Platform ETL Pipeline



Because .bak files use a proprietary binary format exclusive to SQL Server, a two-stage engineering approach was required to bridge legacy infrastructure to a modern open-source environment.

01

Legacy Restoration

The .bak file was restored into a local SQL Server Express instance via SSMS, instantiating the **MarketingAnalytics** relational database.

02

Python Migration Script

A pandas + SQLAlchemy pipeline pulled relational arrays from SQL Server and loaded them into PostgreSQL, dynamically normalizing column names to lowercase to resolve case-sensitivity conflicts.

Cleaning, Auditing & Transformation

Before visualization, production tables were audited in PostgreSQL to guarantee strict data integrity for downstream analysis.



Duplicate Event Isolation

A CTE with a window function partitioned the customer timeline. Rows with `row_num > 1` flagged duplicate transactions requiring filtering to prevent skewed conversion rates.



Categorical Normalization

Channel values like 'socialmedia', 'SOCIALMEDIA', and 'Social Media' were standardized within a production SQL View to ensure consistent grouping.



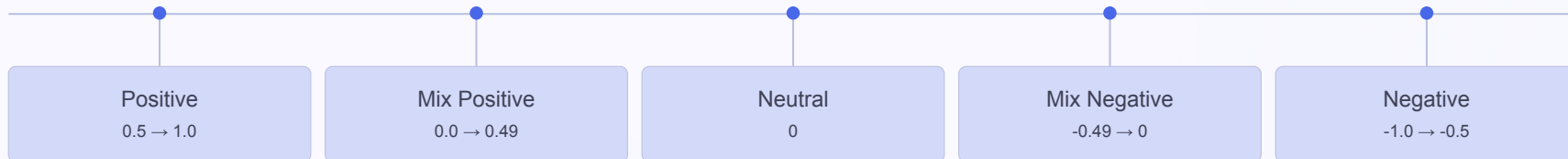
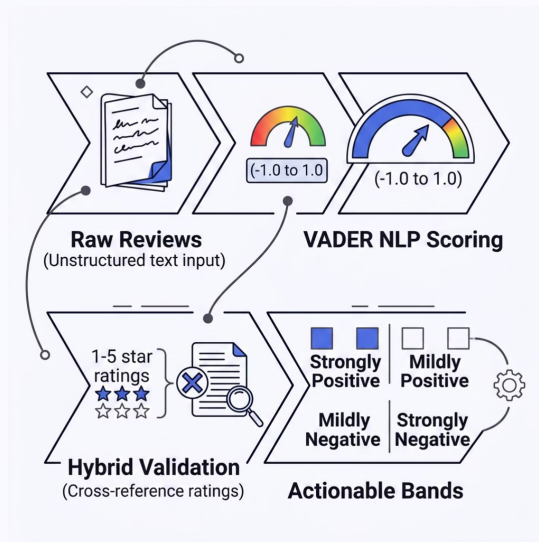
Date Casting

Date fields were explicitly cast to native SQL DATE types rather than strings, eliminating parsing errors upon direct Power BI connection.

Algorithmic Sentiment Analysis

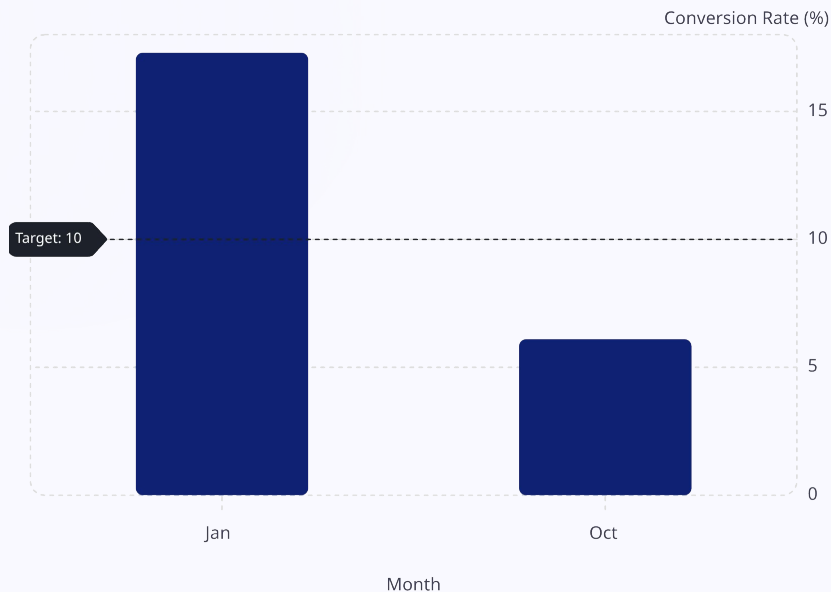
Hybrid Rules-Based Scoring

Rather than relying solely on automated linguistic models, a hybrid classification architecture evaluates textual intent against empirical star ratings — capturing sarcasm, misunderstandings, and rating mismatches.



Key Findings

Conversion Rate Trends



What the Data Shows

Conversion rates fluctuated significantly across 2025. **January peaked at 17.3%** — driven by Ski Boots at 100% conversion — while **October bottomed out at 6.1%**, signaling a clear seasonal gap.



Underperformers: Swim Goggles and Running Shoes recorded conversion rates below 7%, requiring targeted intervention.

Q4 2025 closed at **7.14%** after a strong Q1 rebound, highlighting the need for off-peak promotional strategy.

Engagement Analysis

Reduced Customer Engagement

1.2M+

Peak Views (Jan)

Views declined steadily through 2025, stabilizing near 700K by year-end.

19.57%

Overall CTR

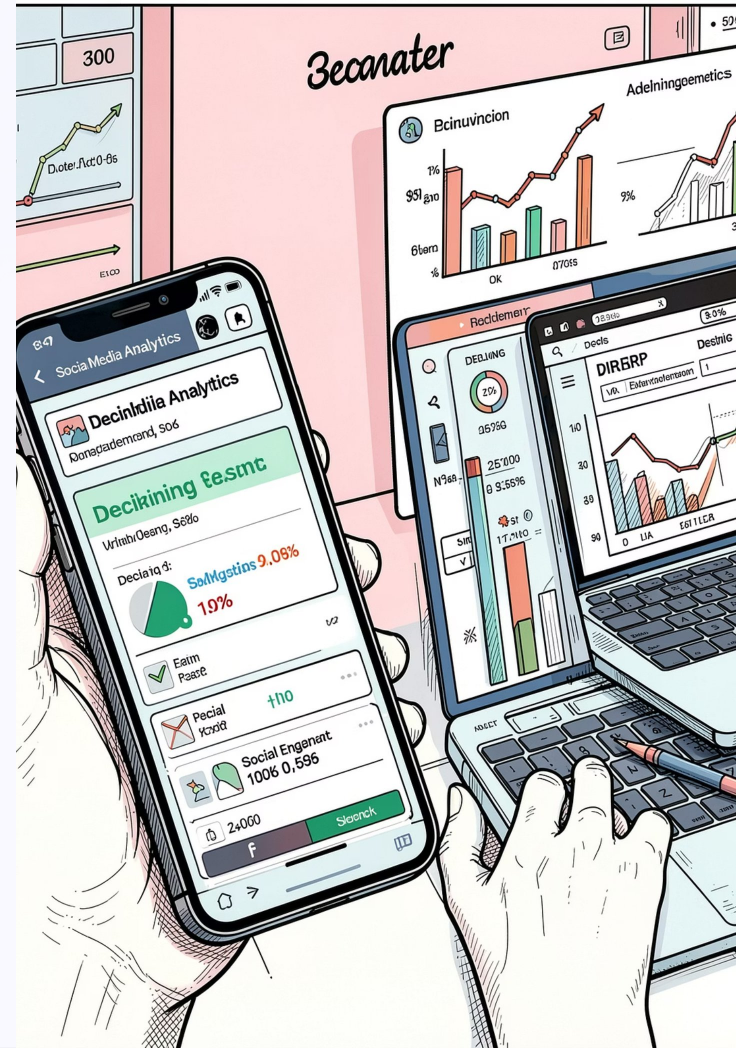
Engaged users still interact effectively, but click-through rate fell from 9.13% (Jan) to 2.75% (Dec).

3.69

Avg Star Rating

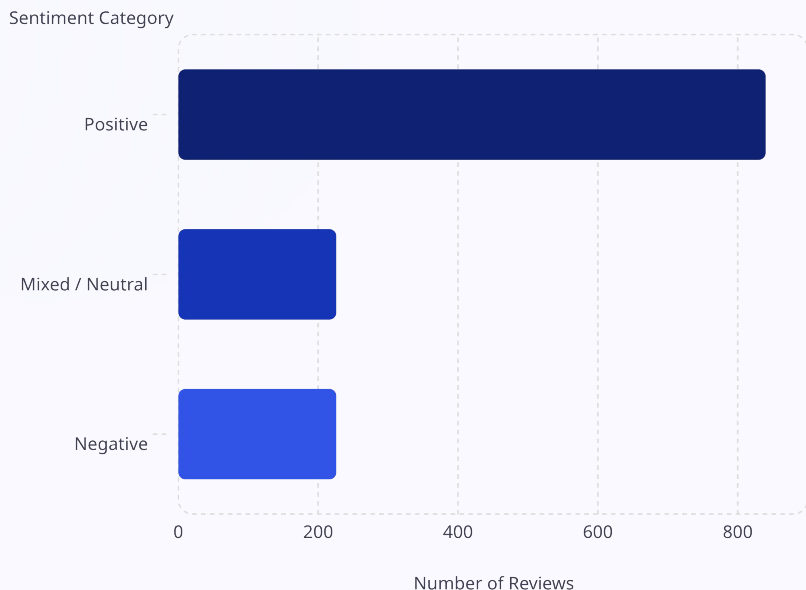
Consistent but below the 4.0 target — products under 3.5 need focused improvement.

Outliers: Soccer Ball and Ice Skates (low-price tier) significantly outperform on clicks — a signal for content strategy replication.



Customer Feedback

Sentiment & Rating Distribution



Reading the Signals

Positive sentiment reviews **outnumber negative ones nearly 4:1**, reflecting a generally satisfied customer base. However, **21%+ mixed/neutral responses** represent a clear conversion opportunity.

- 400+ reviews each at 4 and 5 stars confirm strong product satisfaction
- Only 80 one-star and 153 two-star reviews — a manageable long tail
- Mixed sentiment reviews are the highest-leverage target for satisfaction improvement



Strategic Recommendations

Actionable Goals & Next Steps

1

Increase Conversion Rates

Focus on Hockey Stick, Ski Boots, and Baseball Gloves during peak months (Jan, Sep). Deploy targeted promotions for Swim Goggles and Running Shoes to close the sub-7% gap.

2

Enhance Customer Engagement

Replicate the content strategy behind Soccer Ball and Ice Skates. Optimize CTA placement in social and blog content, especially during Sep–Dec low-engagement windows.

3

Improve Feedback Scores

Implement a structured feedback loop analyzing mixed and negative reviews. Follow up with dissatisfied customers to resolve issues and encourage re-rating toward the 4.0 target.